be drawn as squares, octagons or any kind of polygon, really: depending on these choices, we'll need a different algorithm.

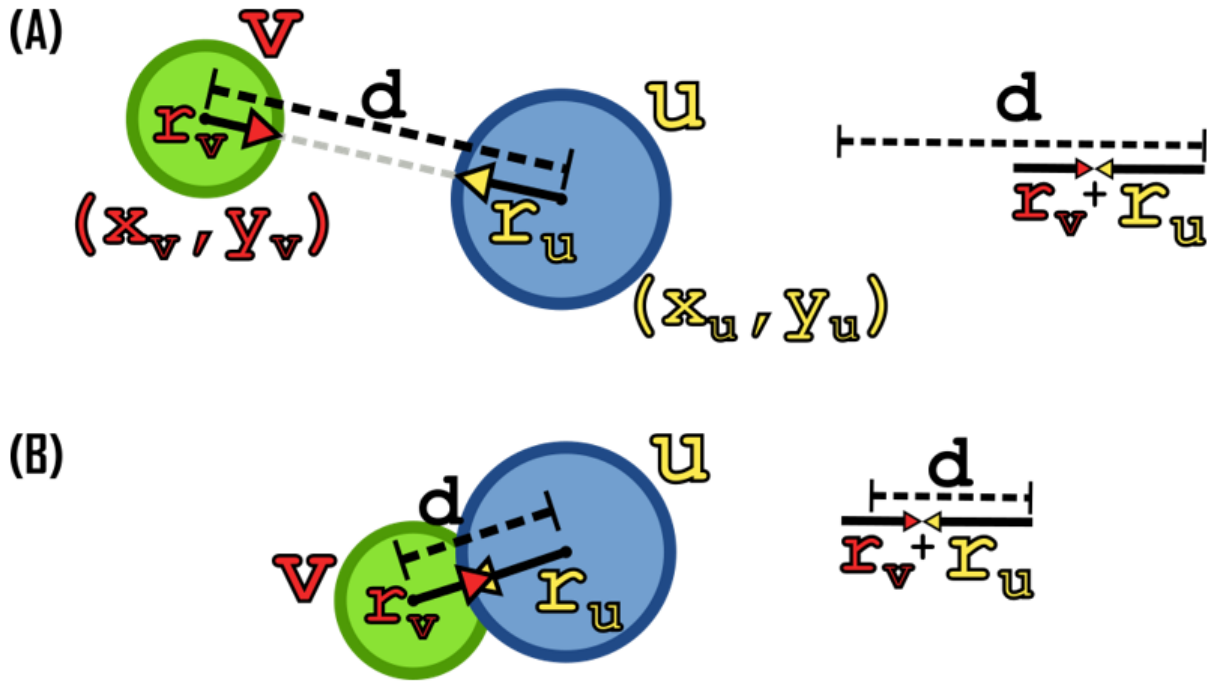


Figure 15.27 Vertex intersection with another vertex. For two circles to overlap, the distance d between their centers must be smaller than the sum of the circles' radii: if $d>r_v+r_u$, like in (A), we are clear; when $d \leq r_v+r_u$, like in (B), the two circles overlap.

Here, we'll assume that vertices are drawn with circles (punctiform vertices being an edge-case, a circle with radius close to zero), and edges are drawn with segments: this will give you the tools to handle the simplest solution, and a basis to build more complex ones, if needed.

For what concerns vertices, in particular, if you choose to use squares or regular polygons, you can always consider the *circumscribed circle*[29] or *minimum bounding circle*[30] of the